

The Program in Action

When you run *dna.py*, the output will look like this:

```
DNA Animation, by Al Sweigart al@inventwithpython.com
Press Ctrl-C to quit...
  #G-C#
  #C---G#
  #T-----A#
  #T-----A#
  #A-----T#
  #G-----C#
  #G---C#
  #C-G#
  ##
  #T-A#
  #C---G#
  #G-----C#
  #G-----C#
  #T-----A#
  #A-----T#
  #C---G#
  #G-C#
  ##
  #T-A#
  #T---A#
  #A-----T#
--snip--
```

How It Works

Similar to Project 15, “Deep Cave,” and Project 20, “Digital Stream,” this program creates a scrolling animation by printing strings from the `ROWS` list. The AT and CG pairs are inserted into each string with the `format()` string method.

```
1. """DNA, by Al Sweigart al@inventwithpython.com
2. A simple animation of a DNA double-helix. Press Ctrl-C to stop.
3. Inspired by matoken https://asciinema.org/a/155441
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: short, artistic, scrolling, science"""
6.
7. import random, sys, time
8.
9. PAUSE = 0.15 # (!) Try changing this to 0.5 or 0.0.
10.
11. # These are the individual rows of the DNA animation:
12. ROWS = [
13.     #123456789 <- Use this to measure the number of spaces:
14.     '      ##', # Index 0 has no {}.
15.     '      #{}-{}#',
16.     '      #{}---{}#',
```